

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer with production experience deploying multi-agent pipelines in regulated environments - HIPAA-conscious healthcare AI, schema-validated inter-agent contracts, audit-ready reasoning traces, and compliance-grade hallucination guardrails. 2.5+ years building AI systems that reason over structured regulatory documents and produce explainable, auditable outputs. Directly transferable to financial compliance automation (AML, KYC, KYB): the same architectural discipline that prevents cascading hallucinations in healthcare claims processing applies to transaction monitoring and identity verification agent design. Master's degree, Kent State University. Builder mindset: multiple independently shipped products.

CORE COMPETENCIES

Multi-Agent Orchestration • Agentic Workflow Design • RAG Systems • Compliance-Grade AI Reliability • Audit-Ready Reasoning Traces • Schema-Validated Agent Contracts • Document Processing / AI Vision • LlamaIndex • LangChain • Python • FastAPI • Docker • Regulated Industry AI (HIPAA) • Production LLM Deployment • Fraud Risk Modeling

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology & AI Consulting (HIPAA-regulated)

- Designed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%. Grounding-rule constraints - not prompt engineering - were the key reliability lever: a production pattern directly applicable to financial compliance agent design.
- Deployed **RAG systems for clinical knowledge retrieval** (LlamaIndex, recursive semantic chunking, transformer embeddings); improved contextual retrieval precision by ~35%, reduced irrelevant retrieval by >30%. Grounded response alignment exceeded 90% in evaluation - compliance-grade reliability in a regulated environment.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for fraud-adjacent risk scoring (patient care engagement, support prioritization); improved recall by 15-20% on high-risk categories while holding precision >90%.
- Packaged ML/LLM inference as **FastAPI/Flask** services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%. Maintained audit-trail documentation for all model decisions in HIPAA-conscious context.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP

- Optimized T-SQL stored procedures for compliance-sensitive financial modules (Synthesis Order-to-Cash - contracts, nominations, allocations, invoicing); reduced query execution time ~20% and latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema updates; reduced deployment errors >30%. Supported **RBAC implementation and audit logging** for financial modules in a compliance-sensitive environment.
- Designed **Grafana performance dashboards** tracking deadlocks, CPU/I/O contention, long-running queries; cut incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics, Applied NLP

- Built transformer-based video summarization system; **60-70% reduction in manual review time**.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~30%, raised contextual accuracy >85%.

KEY PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations - the core reliability pattern for compliance AI. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity for duplicate claim detection; **explainable risk scoring with audit-ready reasoning traces**. Direct architectural parallel to AML/KYC agent design: each agent validates its input before processing, producing tamper-evident decision logs.

LlamaIndex, LangChain, Python, multi-agent orchestration, RAG, vector search, fraud risk modeling

Manga Lens - Multi-Provider AI Document Processing shipped

Chrome extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local) for real-time document content extraction across 29 structured document formats. Provider abstraction layer with per-provider payload optimization, 7-day cache, graceful degradation. Demonstrates production-grade document AI extraction patterns applicable to KYC document processing pipelines.

JavaScript, Claude API, GPT-4o, Ollama, multi-provider AI vision, document extraction

E-Farming Digital Marketplace founder

Independently designed, built, and launched a PHP/MySQL/Bootstrap marketplace; onboarded **80-120 active users** in phase one. Full product ownership from architecture to launch - the same founding-engineer mentality required at Sphinx's stage.

PHP, MySQL, Bootstrap, full-stack product development

Dream Decoder - Multimodal AI Platform with Structured Pipeline Architecture

FastAPI + React app with intermediate structured prompt transformation layers between LLM stages - improved contextual alignment **~30%** and first-pass success rate **~25-30%**. Demonstrates LLM pipeline seam engineering: the same pattern prevents context corruption between compliance agent stages.

FastAPI, React/TypeScript, Perplexity Sonar, Replicate, structured LLM pipeline architecture

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Agent & AI: Multi-agent orchestration, LlamaIndex, LangChain, RAG system design, agentic workflows, schema-validated agent contracts, audit-trail AI, hallucination mitigation, vector search

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (T-SQL, PostgreSQL), PHP

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, YOLOv8

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, structured logging

Domains: Regulated Industry AI (HIPAA-conscious healthcare), Compliance-Grade AI Systems, Fraud Risk Modeling, Production LLM Deployment, Applied AI

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required